CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 8: Remainder and Factor Theorems

Previous year board questions, topic-wise: 14 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 The factor theorem

1. Which factor is common to $x^2 - 4$ and $x^3 - x^2 - 4x + 4$? (a) $(x + 1)$ (b) $(x - 1)$ (c) $(x - 2)$ (d) $(x - 4)$ ICSE 2025, 1 mark

2 Factorising a cubic polynomial

2. $(x - 2)$ is a factor of $2x^3 - 7x^2 + kx - 2$. Find k, and then factorise the polynomial completely. ICSE 2025, 3 marks
3. $(x - 1)$ is a factor of $3x^3 + 8x^2 - 15x + k$. Find k, and then factorise the polynomial completely. ICSE 2024, 3 marks
4. Use the factor theorem to factorise $2x^3 - x^2 - 13x - 6$ completely. ICSE 2023, 4 marks
5. Use the factor theorem to factorise $6x^3 + 17x^2 + 4x - 12$ completely. ICSE 2020, 3 marks
6. Use the factor theorem to show that $(x - 2)$ is a factor of $x^3 + x^2 - 4x - 4$. Then factorise the polynomial completely. ICSE 2019, 3 marks
7. Use the remainder theorem to factorise $2x^3 + 3x^2 - 9x - 10$. ICSE 2018, 3 marks

3 The remainder theorem

8. What must be subtracted from $x^3 + x^2 - 2x + 1$ so that the result is exactly divisible by $(x - 3)$? (a) -31 (b) -30 (c) 30 (d) 31 ICSE 2024, 1 mark
9. What must be added to $2x^3 - 3x^2 - 8x$ so that it leaves a remainder of 10 when divided by $2x + 1$? ICSE 2020, 3 marks
10. Find the remainders when $x^3 + (kx + 8)x + k$ is divided by $x + 1$ and by $x - 2$. The two remainders add up to 1. Find k. ICSE 2019, 3 marks
11. What must be subtracted from $16x^3 - 8x^2 + 4x + 7$ so that the result has $2x + 1$ as a factor? ICSE 2017, 3 marks

4 Finding unknown constants

12. $(x - 2)$ is a factor of $x^3 - kx - 12$. Find k. (a) 3 (b) 2 (c) -2 (d) -3 ICSE 2023, 1 mark
13. Find a if $(x - a)$ is a factor of $3x^3 + x^2 - ax - 81$. ICSE 2023, 4 marks

14. $(x+2)$ and $(x+3)$ are factors of x^3+ax+b. Find a and b. ICSE 2018, 3 marks

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